K-DENSE WEBINAR · SKILLS 101

Build Your Own Scientific Agent Skill

Event by K-Dense Inc.

Host: Yuhuan He · AI Engineer, K-Dense

Wed, Jul 8, 2026 · Live webinar

k-dense.ai

PART 1

Intro

What an Agent Skill is — and why it matters.

WHAT IS AN AGENT SKILL

A small folder that teaches an AI agent something it isn't reliably good at on its own.

Think of it as **expertise you hand the model** — a workflow, a house style, the quirks of a niche package, a domain procedure. The model is already smart; a skill adds the specific knowledge and steps it's missing.

THE LOAD-BEARING PART

The skill's **description** is what makes the agent reach for it at the right moment — it loads on demand, only when your request matches.

💡 **Rule of thumb:** if you keep explaining the same thing to the model over and over, that's a skill waiting to be written.

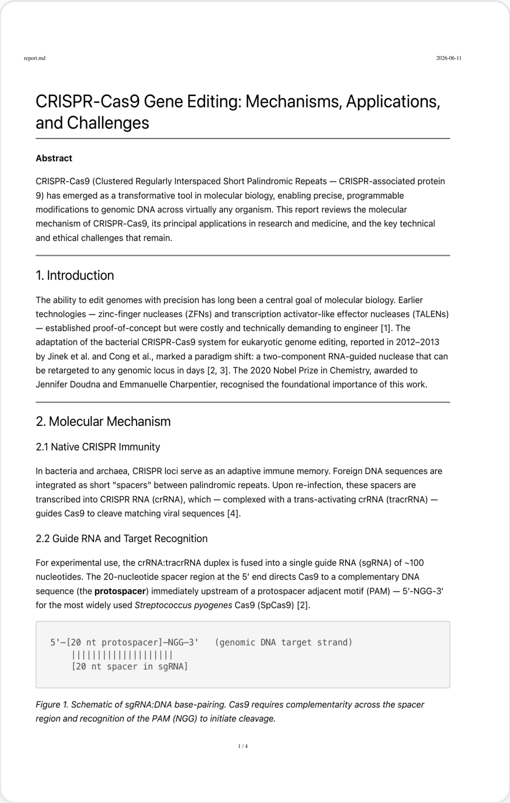
It follows the open Agent Skills standard — the same skill works across Claude Code, Cursor, Codex, and more.

A REAL USE CASE — SAME MODEL, SAME PROMPT

THE PROMPT (IDENTICAL IN BOTH RUNS)

`Write a ~2-page scientific report on CRISPR-Cas9 gene editing, with figures and citations.`

Without a skill



report.md 2026-06-11

CRISPR-Cas9 Gene Editing: Mechanisms, Applications, and Challenges

Abstract

CRISPR-Cas9 (Clustered Regularly Interspaced Short Palindromic Repeats — CRISPR-associated protein 9) has emerged as a transformative tool in molecular biology, enabling precise, programmable modifications to genomic DNA across virtually any organism. This report reviews the molecular mechanism of CRISPR-Cas9, its principal applications in research and medicine, and the key technical and ethical challenges that remain.

1. Introduction

The ability to edit genomes with precision has long been a central goal of molecular biology. Earlier technologies — zinc-finger nucleases (ZFNs) and transcription activator-like effector nucleases (TALENs) — established proof-of-concept but were costly and technically demanding to engineer [1]. The adaptation of the bacterial CRISPR-Cas9 system for eukaryotic genome editing, reported in 2012–2013 by Jinek et al. and Cong et al., marked a paradigm shift: a two-component RNA-guided nuclease that can be retargeted to any genomic locus in days [2, 3]. The 2020 Nobel Prize in Chemistry, awarded to Jennifer Doudna and Emmanuelle Charpentier, recognised the foundational importance of this work.

2. Molecular Mechanism

2.1 Native CRISPR Immunity

In bacteria and archaea, CRISPR loci serve as an adaptive immune memory. Foreign DNA sequences are integrated as short "spacers" between palindromic repeats. Upon re-infection, these spacers are transcribed into CRISPR RNA (crRNA), which — complexed with a trans-activating crRNA (tracrRNA) — guides Cas9 to cleave matching viral sequences [4].

2.2 Guide RNA and Target Recognition

For experimental use, the crRNA:tracrRNA duplex is fused into a single guide RNA (sgRNA) of ~100 nucleotides. The 20-nucleotide spacer region at the 5′ end directs Cas9 to a complementary DNA sequence (the **protospacer**) immediately upstream of a protospacer adjacent motif (PAM) — 5′-NGG-3′ for the most widely used *Streptococcus pyogenes* Cas9 (SpCas9) [2].

```
5'—[20 nt protospacer]—NGG—3'   (genomic DNA target strand)
    ||||||||||||||||||||
    [20 nt spacer in sgRNA]
```

Figure 1. Schematic of sgRNA:DNA base-pairing. Cas9 requires complementarity across the spacer region and recognition of the PAM (NGG) to initiate cleavage.

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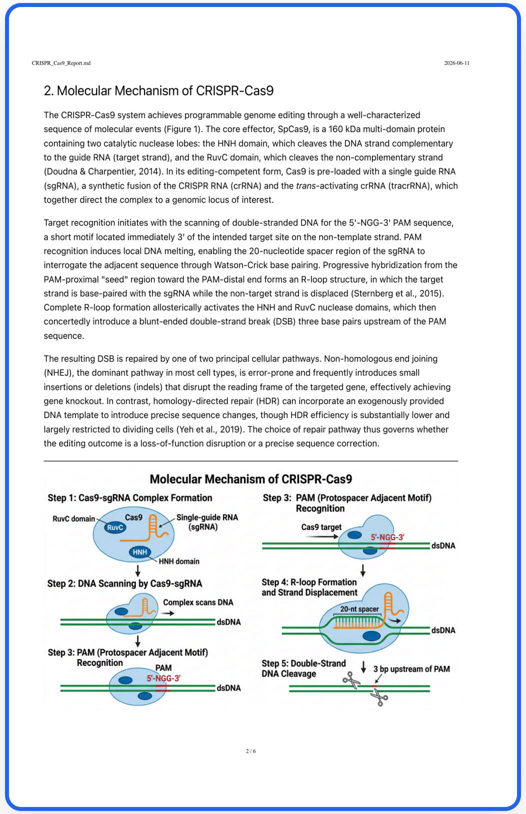
Plain prose · only an ASCII text schematic

With `scientific-writing`



CRISPR_Cas9_Report.md 2026-06-11

2. Molecular Mechanism of CRISPR-Cas9

The CRISPR-Cas9 system achieves programmable genome editing through a well-characterized sequence of molecular events (Figure 1). The core effector, SpCas9, is a 160 kDa multi-domain protein containing two catalytic nuclease lobes: the HNH domain, which cleaves the DNA strand complementary to the guide RNA (target strand), and the RuvC domain, which cleaves the non-complementary strand (Doudna & Charpentier, 2014). In its editing-competent form, Cas9 is pre-loaded with a single guide RNA (sgRNA), a synthetic fusion of the CRISPR RNA (crRNA) and the *trans*-activating crRNA (tracrRNA), which together direct the complex to a genomic locus of interest.

Target recognition initiates with the scanning of double-stranded DNA for the 5'-NGG-3' PAM sequence, a short motif located immediately 3' of the intended target site on the non-template strand. PAM recognition induces local DNA melting, enabling the 20-nucleotide spacer region of the sgRNA to interrogate the adjacent sequence through Watson-Crick base pairing. Progressive hybridization from the PAM-proximal "seed" region toward the PAM-distal end forms an R-loop structure, in which the target strand is base-paired with the sgRNA while the non-target strand is displaced (Sternberg et al., 2015). Complete R-loop formation allosterically activates the HNH and RuvC nuclease domains, which then concertedly introduce a blunt-ended double-strand break (DSB) three base pairs upstream of the PAM sequence.

The resulting DSB is repaired by one of two principal cellular pathways. Non-homologous end joining (NHEJ), the dominant pathway in most cell types, is error-prone and frequently introduces small insertions or deletions (indels) that disrupt the reading frame of the targeted gene, effectively achieving gene knockout. In contrast, homology-directed repair (HDR) can incorporate an exogenously provided DNA template to introduce precise sequence changes, though HDR efficiency is substantially lower and largely restricted to dividing cells (Yeh et al., 2019). The choice of repair pathway thus governs whether the editing outcome is a loss-of-function disruption or a precise sequence correction.

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Real multi-panel figures · named citations · sources folder

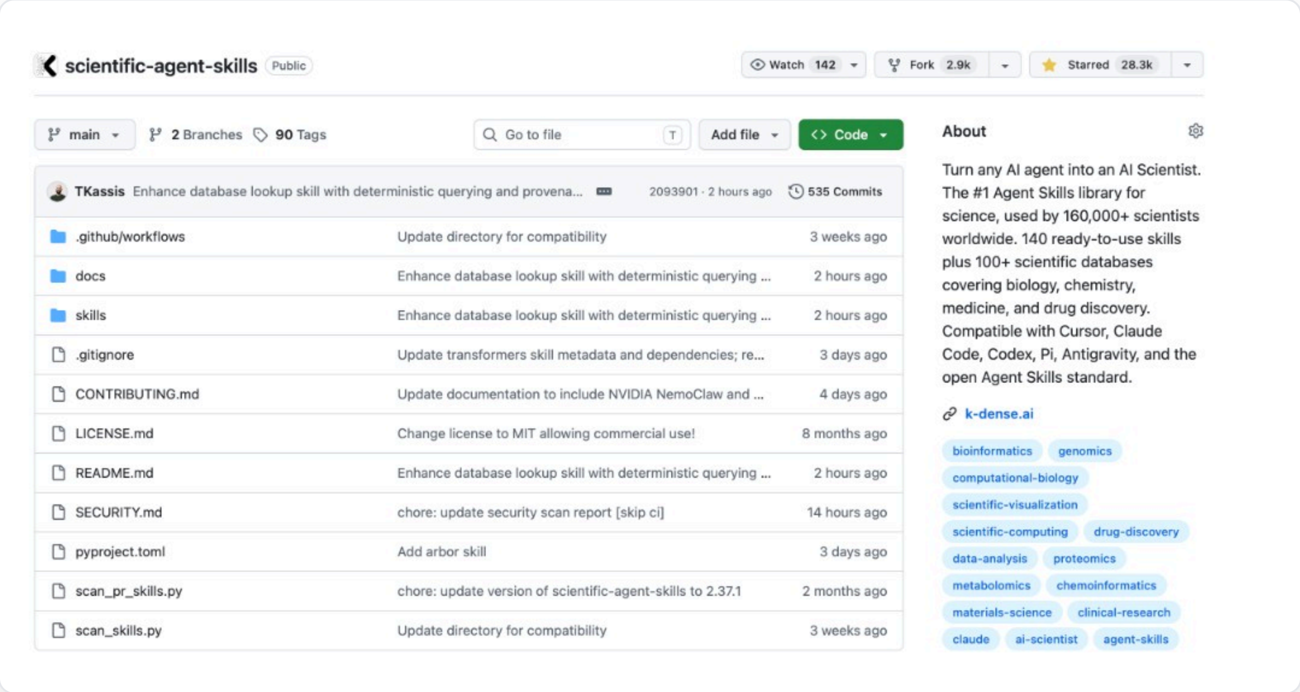
Built with K-Dense's `scientific-writing` skill — plus its companions `research-lookup`, `scientific-schematics`, and `generate-image`. Click a page to open the full report.

TRY IT YOURSELF

A whole open library of these

The report you just saw used four skills from K-Dense's **open, MIT-licensed** registry — and there are many more, spanning the sciences. **Repo:** github.com/K-Dense-AI/scientific-agent-skills

135+	250+	60+
Scientific Agent Skills	**Databases & Data Sources**	**Optimized Python Packages**
Reviewed by domain scientists	PubMed, ChEMBL, cBioPortal, gnomAD...	Scanpy, RDKit, OpenMM, PyMC...



HOW IT WORKS

How Agent Skills Work

PICKED AUTOMATICALLY, ON DEMAND

Discovery → **Activation** → **Execution**

Discovery
Agent reads each skill's name & description

Activation
The matching skill loads into context

Execution
Agent follows the skill's instructions

WHAT'S INSIDE THE FOLDER

my-skill/

- **SKILL.md** Required — instructions + metadata
- **references/** Optional — docs to read
- **scripts/** Optional — code to run
- **assets/** Optional — templates to use

Each skill is just a folder with a `SKILL.md` file — easy to read, edit, share, and audit.

WHAT MAKES A SKILL GOOD

Three things — and what a `SKILL.md` looks like

A description that triggers

Third person, packed with the words a user would actually type.

Teach only what's missing

Don't wrap what the model already knows. Target the gaps.

Keep it concise

Lean SKILL.md; push detail into `references/` when needed.

```
---
name: plain-language-summary
description: Rewrite dense or technical text into a clear plain-language summary
with a fixed structure (TL;DR, why it matters, key points, caveats). Use when the
user asks to summarize, simplify, explain, "put this in plain language", or make
something readable for a general / non-expert audience.
---

## When to use
Any time the user wants technical, jargon-heavy, or dense material made clear for a
general reader — abstracts, documentation, reports, emails, papers.

## Output format (ALWAYS use exactly this)
**TL;DR:** one plain sentence anyone could understand.
**Why it matters:** 1–2 sentences on the real-world significance.
**Key points:**
- 3–5 bullets, each jargon-free and self-contained.
**Caveats:** what's uncertain, assumed, or out of scope (1–2 bullets).

## Rules
- No jargon without a 3-word plain explanation in parentheses.
- Prefer everyday words; cut filler.
- Keep the whole thing under ~200 words unless the user asks for more.
- Never invent facts that aren't in the source text; if something is unclear, say so
in Caveats.
```

PART 2

Environment Setup

Get Claude Code and the workshop pack ready.

Hands-on — follow along on your laptop

ENVIRONMENT SETUP

What to have ready

🤖 **Claude Code**

Signed in with a Claude Pro or Max plan. Needs Node.js 18+.

💻 **An editor you like**

e.g. VS Code — run `claude` in its integrated terminal so you can edit, run, and watch files in one window.

```
# 1 · make a working folder (run claude here)
mkdir -p ~/Desktop/skills-workshop
cd ~/Desktop/skills-workshop

# 2 · check Claude Code + skills folder
claude --version
mkdir -p ~/.claude/skills

# 3 · launch Claude Code
claude
```

ENVIRONMENT SETUP

The workshop pack

Download & unzip it so its contents sit **directly inside your working folder**.

```
~/Desktop/skills-workshop/
├── README.md
├── example_skill/
│   └── plain-language-summary/
│       └── SKILL.md                # warm-up skill
├── sample_inputs/                  # text to paste while testing
│   ├── abstract.txt
│   ├── experiment_notes.txt
│   ├── kdense_blog.txt
│   ├── messy_meeting_notes.txt
│   └── results.csv
├── templates/
│   ├── paper-draft-SKILL.md        # open-build skill
│   └── your-skill-template.md      # blank skill to copy
└── use_case/                       # before/after PDFs (intro)
    ├── CRISPR_no_skill.pdf
    └── CRISPR_with_skill.pdf
```

All sample inputs are synthetic or public — safe to paste.

PART 3

The Goal

What we're going to build together.

THE GOAL

Build your own Agent Skill — and prove it changed the output.

By the end you'll have built a skill from scratch, installed it in Claude Code, and run the **same prompt with it on and off** to see the difference.

Build	Install	Prove it
Write a `SKILL.md` from scratch	Drop it into Claude Code & restart	Show the before / after

Go further (reach): build a more complex skill with a `references/` file, then write a tiny eval scoring the output with vs. without it.

PART 4

Warm-up

Install a ready-made skill and feel the difference.

Hands-on — follow along on your laptop

PART 5

Open Build

Build your own skill from scratch — then prove it helped.

Hands-on — follow along on your laptop

OPEN BUILD · YOUR TURN

Now do your own — the build loop

1. **Baseline** — run your prompt with no skill, keep it
2. **Make the folder** — lowercase, hyphens = skill name
3. **Write** `SKILL.md` — a load-bearing description
4. **Write the body** — the exact output you want
5. **Install + restart** Claude Code
6. **Test it triggers** on the same prompt

Copy the blank `templates/your-skill-template.md` and swap in your own idea.

7 · ADD A TINY EVAL

Check	Off	On
Used the IMRAD structure?	—	—
No numbers beyond the abstract?	—	—
No invented citations?	—	—
Listed the gaps to fill?	—	—

Score the output with vs. without your skill — run the same prompt **off** then **on**, and fill in a short rubric you design.

Skill ideas if you're stuck: plain-language paper summary · methods-paragraph writer · lab-protocol checklist · weekly research update · a figure/plot house-style · a niche-package quickstart · meeting-notes formatter · commit-message style · code-review checklist.

REACH · GO FURTHER

Make it bigger: `references/` · `scripts/` · `assets/`

A simple skill is one `SKILL.md`. Bigger skills add three optional folders — the difference is **what the agent does with each**.

`references/`

The agent *reads* it

Markdown docs loaded only when needed — e.g. your lab's section conventions.

`scripts/`

The agent *runs* it

Helper code for fragile / exact steps — call an API, parse a file.

`assets/`

The agent *uses* it

Templates, styles, sample data it copies into outputs.

💡 **Example:** for `paper-draft`, add a `references/imrad.md` with your target journal's section conventions (word limits, Methods vs. Results, citation style), then point to it from `SKILL.md` — Claude reads it only when needed.

OPEN BUILD · DEMOS

Show & tell

Each group, **5–8 minutes** — no slides needed, just screen-share.

Your skill

What it does and who it's for.

The before / after

Same prompt, with vs. without it — share both outputs.

One surprise

A description tweak that fixed triggering, a failure, an idea it unlocked.

PART 6

Wrap-Up

Where to go next — and how to stay in touch.

RESOURCES

Links & further reading

Claude Code — install & docs
`code.claude.com/docs/en/overview`

Claude Code — Skills docs
`code.claude.com/docs/en/skills`

Agent Skills — open standard
`agentskills.io`

Anthropic — skill-authoring guide
`claude.com/docs/skills/how-to`

Example science skills (inspiration)
`github.com/K-Dense-AI/scientific-agent-skills`

Workshop pack (download)
`drive.google.com → skills-workshop`

JOIN THE CONVERSATION

Connect With Us

These skills come from **K-Dense** — our AI co-scientist for autonomous research. The skills are all open source; the product is where it all comes together.

Try K-Dense `app.k-dense.ai` →

Scan → our open skills repo
github.com/K-Dense-AI

GitHub
K-Dense-AI

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@k_dense_ai

YouTube
@K-Dense-Inc

LinkedIn
K-Dense Inc